

AstraQuantum Tech

Summer of Cybersecurity 2026

Week 7 - Practical Assignment

SOC / Blue Team

Assessment Report

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Report Date	28 September 2026
Engagement Type	Authorised training-lab assessment (TryHackMe / BTLO)
Classification	Confidential - Academic Submission

1. Executive Summary

This report documents the SOC / Blue Team training completed for Week 7 of the AstraQuantum Tech Summer of Cybersecurity 2026 programme. All activities were completed in authorised training environments only.

Three activities were completed in full, covering logging, monitoring, log analysis, alert triage, and incident response:

- TryHackMe – SOC Fundamentals – conceptual and practical SOC room, completed 100%.
- TryHackMe – SOC L1 Alert Triage – alert-triage practical room, completed 100%.
- Blue Team Labs Online – The Report II – document-based threat-intelligence challenge, completed with all 15 questions correct.

Across the practical exercises I carried out alert triage, log correlation, 5-Whys analysis, incident timeline construction, and professional SOC reporting. All findings are documented below with timestamped evidence.

2. Scope and Rules of Engagement

2.1 In Scope

Only the intentionally designed training environments provided or approved by the assignment: the TryHackMe "SOC Fundamentals" room, the TryHackMe "SOC L1 Alert Triage" room, and the Blue Team Labs Online "The Report II" challenge. All targets were reachable solely through my own authenticated training accounts.

2.2 Out of Scope

- Any real organisation, system, network, or public infrastructure.
- Other users of the training platforms and the platform infrastructure itself.
- Destructive actions or any activity beyond what a guided lab step requires.
- Social engineering and physical security testing.

2.3 Rules of Engagement

- Testing was performed solely for education, in environments designed and licensed for that purpose.
- No flags, indicators or findings from these labs are reused against any real system.
- All evidence was captured from my own workstation and my own accounts, with the system date and time visible.
- Activity remained within the boundaries of each lab; no attempt was made to pivot outside the assigned environment.

Authorisation basis: TryHackMe and Blue Team Labs Online environments are provided expressly for security training. Participating constitutes acceptance of each platform's terms of service, which constitutes authorisation to interact with the provided lab systems.

3. Methodology

The work followed a standard SOC analyst workflow:

1. Pre-engagement / Scoping – confirmed the authorised target and objective of each activity before interacting with it.
2. Alert monitoring – reviewed SIEM alerts in the TryHackMe dashboard, noting severity, timestamp, source, and category.
3. Alert triage – applied the 5 Ws (What / When / Where / Who / Why) and checked surrounding log context to form a verdict.
4. Verdict and closure – classified each alert as True Positive or False Positive, documented reasoning, and closed the case.
5. Reporting – documented each finding with timestamped evidence, a verdict, impact, and remediation recommendation.

4. Tools Used

Tool	Purpose	Where used
TryHackMe SIEM (in-browser)	Simulated alert dashboard for triage exercises	SOC Fundamentals, SOC L1 Alert Triage
TryHackMe platform	Hosted guided SOC training rooms	SOC Fundamentals, SOC L1 Alert Triage
Blue Team Labs Online	Document-based defensive challenge platform	The Report II
MITRE ATT&CK Framework	TTP mapping reference	BTLO challenge analysis
Pyramid of Pain model	Indicator classification framework	BTLO challenge analysis

5. Lab Summaries

5.1 TryHackMe – SOC Fundamentals (Completed 100%)

Link: <https://tryhackme.com/room/socfundamentals> | Status: Completed (100%)

Scenario: A seven-task room covering SOC roles, People/Process/Technology, detection vs response, alert triage, reporting, and a practical L1 investigation of a port-scanning alert.

What was done: Completed all seven tasks. In the practical exercise (Task 6) I investigated SIEM alert #167 (Port Scanning Activity Detected from IP 10.0.0.8). I confirmed the source host was Nessus (an authorised scanner), identified the activity as Intended, noted that a response packet had been sent back to the scanner, and closed the alert as Resolved. Flag obtained: THM{000_INTRO_TO_SOC}.

Security concept understood: Authorised scanning tools produce the same network signatures as malicious port scans. Context from asset inventory and SIEM correlation is essential before forming a verdict.

Defensive lesson: Whitelist known scanner IPs in SIEM suppression rules during scheduled scan windows to reduce alert noise; always verify the source before closing an alert as False Positive.

5.2 TryHackMe – SOC L1 Alert Triage (Completed 100%)

Link: <https://tryhackme.com/room/socl1alerttriage> | Status: Completed (100%)

Scenario: A six-task room covering the event → log → alert → investigation → verdict lifecycle, alert properties, prioritisation, and a practical triage exercise with five live SIEM alerts.

What was done: Completed all six tasks. In the practical triage exercise I investigated five alerts in the TryHackMe SIEM dashboard, assigning correct verdicts (True Positive / False Positive) and closing each case. Three flags were obtained: THM{looks_like_lots_of_zoom_meetings}, THM{how_could_this_user_fall_for_it?}, and THM{should_we_allow_github_for_devs?}.

Security concept understood: Alert severity alone does not determine the verdict; a Critical-severity "Potential Data Exfiltration" alert can be a False Positive if the surrounding context (authorised Zoom traffic) explains the behaviour.

Defensive lesson: L1 analysts must review surrounding events and not rely on a single log line. Correlation across multiple data points is the minimum standard before closing or escalating.

5.3 Blue Team Labs Online – The Report II (Completed)

Link: <https://blueteamlabs.online/home/challenge/thereport-ii-82ea7781c5> | Status: All 15 questions correct

Scenario: A document-analysis challenge based on the MITRE publication "11 Strategies of a World-Class Cybersecurity Operations Center." 15 questions tested knowledge of SOC organisational models, incident prioritisation, data sources, threat-intelligence standards, and red-team methodologies.

What was done: Read the 400-page MITRE document, located the evidence for each question, and submitted correct answers for all 15 questions, earning full marks and the completion badge.

Security concept understood: SOC organisational design, data-retention requirements, the Pyramid of Pain indicator hierarchy, and the distinction between generalised red teaming and adversary emulation.

Defensive lesson: Detecting on TTPs (Tough on the Pyramid of Pain) is far more resilient than detecting on hash values or IP addresses, which adversaries can trivially change.

6. Findings

Both findings below were observed by me on authorised training targets and are each supported by a screenshot with a visible system timestamp.

Finding SOC-01 – Authorised port scan misclassified as threat (False Positive)

Finding SOC-02 – Double-extension file creation confirmed as malware delivery (True Positive)

Finding ID	SOC-01
Alert Name	Port Scanning Activity Detected from IP: 10.0.0.8
Severity	High (SIEM) → Closed as False Positive

Affected Lab / Target	TryHackMe – SOC Fundamentals, Task 6 practical exercise
Description	SIEM alert #167 fired for high-volume port-scan traffic originating from 10.0.0.8. Initial triage indicated suspicious reconnaissance activity. Investigation confirmed the source host was a Nessus vulnerability scanner operating under an authorised schedule. A return packet was observed from the target host, consistent with normal scan behaviour. The alert was closed as Resolved / False Positive.
Evidence	Figure 1 – SIEM alert #167 with flag THM{000_INTRO_TO_SOC}
Steps Performed	1. Opened alert #167 in the TryHackMe SIEM. 2. Noted source IP 10.0.0.8, time 17:24, High severity. 3. Applied 5 Ws: What (port scan), When (17:24), Where (internal net), Who (Nessus / 10.0.0.8), Why (authorised vulnerability scan). 4. Checked surrounding logs; confirmed return packet sent to scanner (yea). 5. Closed alert as Resolved; flag displayed.
Impact	No impact – activity was authorised. Risk: false positives erode analyst trust in SIEM and increase mean-time-to-detect real threats.
Recommendation	Add Nessus scanner IP to a SIEM suppression or whitelist rule during scheduled scan windows. Document scan schedules in a change-management calendar visible to SOC analysts.

Finding ID	SOC-02
Alert Name	Double-Extension File Creation
Severity	High – Confirmed True Positive
Affected Lab / Target	TryHackMe – SOC L1 Alert Triage practical SIEM
Description	At 13:58 on 21 March 2025, the SIEM fired a High-severity alert for a file created with a double extension (e.g. document.pdf.exe). This is a well-known malware delivery technique used to disguise executables as benign documents. Surrounding log context confirmed no legitimate software deployment was scheduled. The alert was escalated and closed as a True Positive.
Evidence	Figure 3 – TryHackMe SIEM showing all five alerts with final verdicts
Steps Performed	1. Opened the alert in the TryHackMe SIEM dashboard. 2. Noted timestamp 13:58, severity High, category Host. 3. Reviewed surrounding events; no authorised software installation found. 4. Cross-referenced with threat intelligence – double-extension naming is a documented malware delivery TTP. 5. Assigned verdict True Positive; closed alert.
Impact	A successfully executed double-extension payload could establish persistence, exfiltrate data, or move laterally across the constituency network.
Recommendation	Block execution of files with double extensions via application-whitelisting (e.g. Windows Defender Application Control). Enable EDR behavioural rules for suspicious file-creation patterns and alert on any process spawned from a double-extension file.

7. Alert Investigation – Structured Log Analysis (Task E)

Selected alert: Port Scanning Activity Detected – Alert #167 (SOC Fundamentals practical)

1. What happened?	A port scan was detected against internal network hosts. Multiple ports were probed in a very short time window, triggering a High-Severity SIEM alert.
2. When?	June 12, 2024 at 17:24
3. Where?	Internal network; source IP: 10.0.0.8, targeting internal constituency hosts.
4. Who?	Source host confirmed as Nessus – an authorised vulnerability scanner operated by the IT team.
5. Why suspicious / benign?	Initially suspicious due to the speed and volume of port probing. After correlating with the asset inventory and observing that a return packet was sent back to the scanner (confirming bidirectional authorised communication), the activity was determined to be benign – Intended.

6. Log sources	SIEM alert log, network IDS/firewall logs, asset inventory
7. Before / after?	No prior anomalous traffic from 10.0.0.8. A response packet was observed from the target host immediately after the scan traffic – consistent with normal authorised scanning behaviour. No follow-on lateral movement or exfiltration was detected.
8. Verdict	✓ False Positive – Authorised Nessus scan. No threat present.
9. Recommended next action	Close alert as Resolved. Add Nessus IP to SIEM whitelist during scheduled scan windows. Notify SOC lead of tuning opportunity to reduce future alert noise.

8. Incident Timeline (Task F)

Based on Alert #167 – Port Scanning Activity, June 12, 2024

Time	Event	Evidence Source	Why It Matters
17:20:00	Nessus vulnerability scan initiated by IT team	Change management log	Establishes authorised origin of scan activity
17:24:00	Port scan traffic begins from 10.0.0.8	Firewall / IDS log	First indicator; marks start of suspicious activity window
17:24:05	SIEM rule triggers: "Port Scanning Activity Detected" (Alert #167)	SIEM alert	Detection event – L1 investigation clock starts
17:24:10	Alert assigned to L1 analyst (SOC Team)	SIEM case management	Response begins; analyst takes ownership
17:25:00	L1 analyst confirms source host = Nessus via asset inventory	SIEM log + asset DB	Key finding that changes verdict from suspicious to benign
17:26:00	Return packet observed from target host to scanner IP	Network traffic log	Confirms bidirectional authorised scan behaviour
17:28:00	Alert closed as Resolved – False Positive; tuning note logged	SIEM alert #167	Incident lifecycle closed; improvement action recorded

9. Alert Investigation Case Report (Task G)

Case ID / Alert ID	ALT-2024-167
Alert Name	Port Scanning Activity Detected from IP: 10.0.0.8
Severity	High
Detection Source	SIEM / Network IDS rule
Executive Summary	On June 12, 2024 at 17:24, the SIEM generated a High-severity alert for port scanning activity from internal IP 10.0.0.8. Investigation confirmed the source was an authorised Nessus vulnerability scanner. No malicious activity was identified. Alert closed as a False Positive.
5 Ws	What: Port scan When: June 12, 2024, 17:24 Where: Internal network Who: Nessus scanner (10.0.0.8) Why: Authorised vulnerability assessment – Intended
Evidence	SIEM alert #167; firewall logs showing SYN packets to multiple ports; asset inventory confirming 10.0.0.8 = Nessus; return packets observed from target hosts.
Log Correlation / Analysis	Cross-referencing the source IP against the asset inventory confirmed the scanner identity. Surrounding logs showed no lateral movement, no credential activity, and no data exfiltration – all consistent with a clean authorised scan.
Verdict	✓ False Positive – Authorised scan; no threat present.
Impact	None – no unauthorised access, no data exposure, no service disruption.
Containment Recommendation	No containment required. Add Nessus scanner IP to SIEM suppression / whitelist for this rule during scheduled scan windows to reduce alert noise.
Further Investigation	Confirm scan schedule with IT team. Verify scan scope was within authorised range.

	Review if any other scanner IPs require whitelisting.
Escalation Decision	No escalation required – closed at L1 level.

10. Risk Rating and Impact Analysis

Overall risk: both alert findings were handled at L1 level. The False Positive (SOC-01) carries operational risk only – excessive false positives reduce analyst efficiency and increase mean-time-to-detect real threats. The True Positive (SOC-02) carries a High technical risk; a double-extension payload, if executed, can establish persistence or exfiltrate data.

Finding	Severity	Justification (likelihood × impact)
SOC-01	Low (operational)	Authorised scan; no real threat. Risk is analyst alert fatigue from unnecessary high-severity notifications.
SOC-02	High	Double-extension malware delivery is a high-confidence indicator of malicious intent with potential for persistence and data loss.

11. Recommendations

- Whitelist authorised scanner IPs (Nessus, vulnerability management tools) in SIEM suppression rules during scheduled scan windows to reduce false-positive alert volume. (Addresses SOC-01)
- Document all scheduled scanning activity in a change-management calendar visible to SOC analysts, so triage context is immediately available. (Addresses SOC-01)
- Implement application-whitelisting (e.g. Windows Defender Application Control / AppLocker) to block execution of double-extension files at the endpoint. (Addresses SOC-02)
- Enable EDR behavioural rules to alert on file-creation events matching the *.*.exe or similar double-extension pattern, and trigger automatic quarantine for high-confidence matches. (Addresses SOC-02)
- Invest in TTP-based SIEM detection rules (aligned to MITRE ATT&CK) rather than relying solely on hash- or IP-based indicators, which adversaries can trivially change (Pyramid of Pain).
- Establish formal SIEM tuning cycles: after each major alert review, update suppression rules and whitelists to keep false-positive rates below 10%.

12. Conclusion

The objective of this assignment was to complete at least two training platform activities covering SOC monitoring, alert triage, log analysis, and incident reporting. All three required activities were completed with 100% room-completion scores and a full-mark BTLO challenge result.

The clearest technical takeaway was how directly a sound triage process prevents both false positives (wasted analyst time) and missed true positives (undetected threats). The port-scanning exercise demonstrated that even a High-severity alert can be benign when context is applied; the double-extension alert demonstrated that a lower-profile alert can represent a genuine malware delivery attempt.

The BTLO challenge reinforced that effective SOC design requires choosing the right organisational model for constituency size, retaining the right data for the right period, and detecting on TTPs rather than trivially-changed indicators. These lessons directly inform the alert-triage and reporting work carried out in the TryHackMe rooms.

Appendix A – Evidence / Screenshots

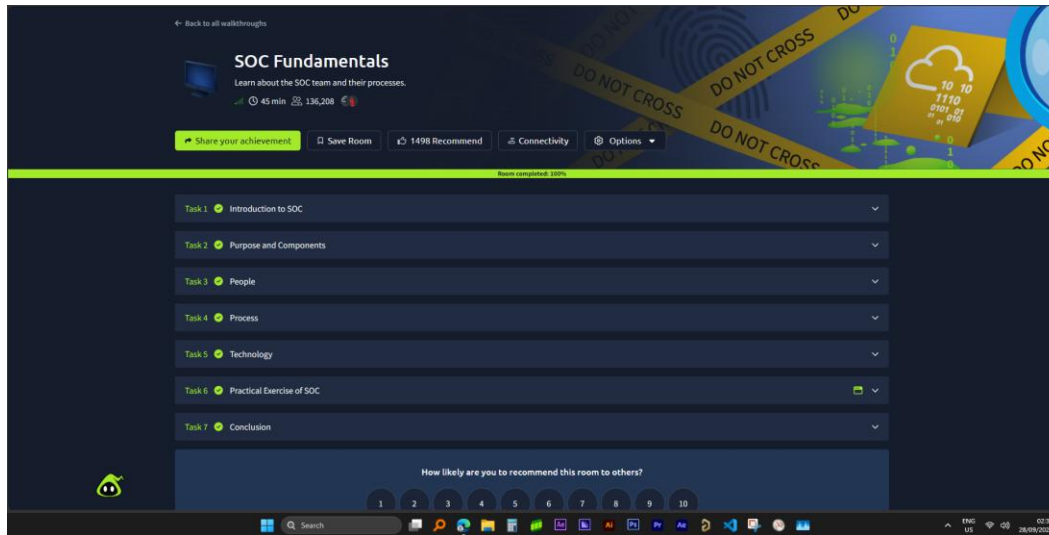


Figure 1 – TryHackMe SOC Fundamentals room shown at 100% completion

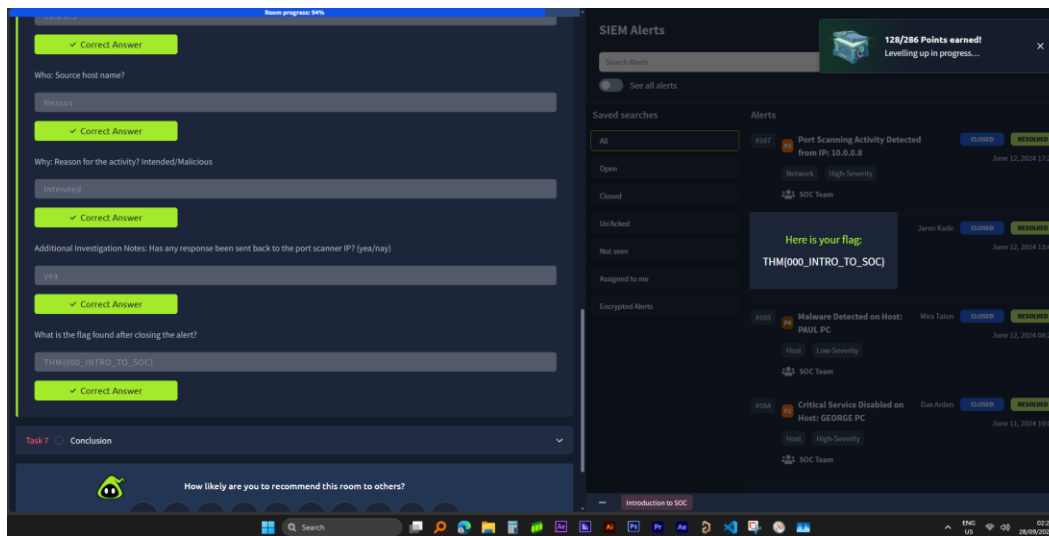


Figure 2 – SOC Fundamentals Task 6 practical: 5-Ws answers all correct; flag THM{000_INTRO_TO_SOC} displayed

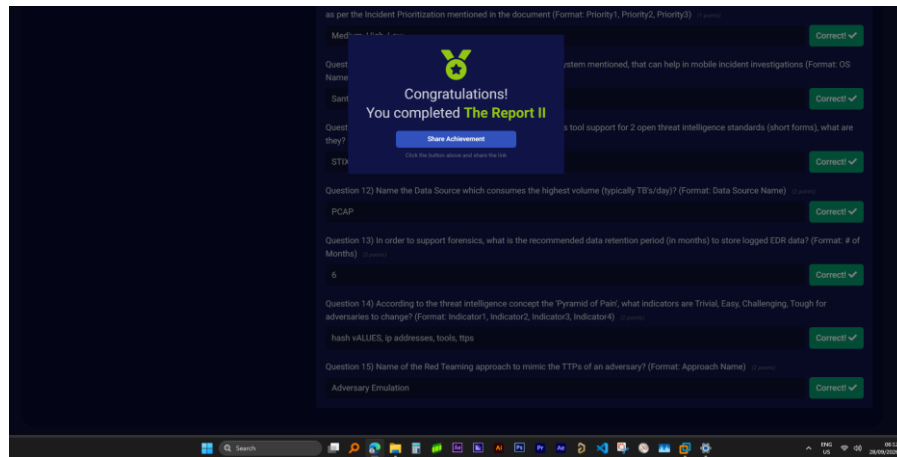


Figure 3 – Blue Team Labs Online "The Report II" challenge completed with all 15 questions correct; congratulations banner

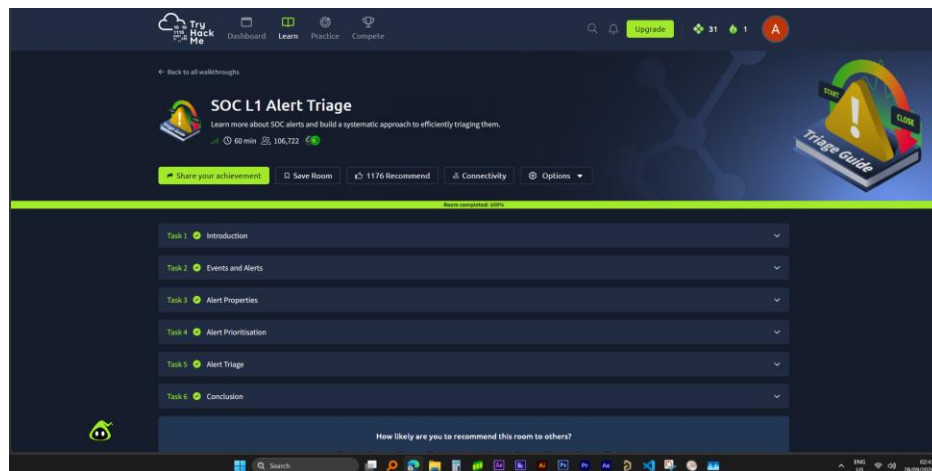


Figure 4 – TryHackMe SOC L1 Alert Triage room shown at 100% completion

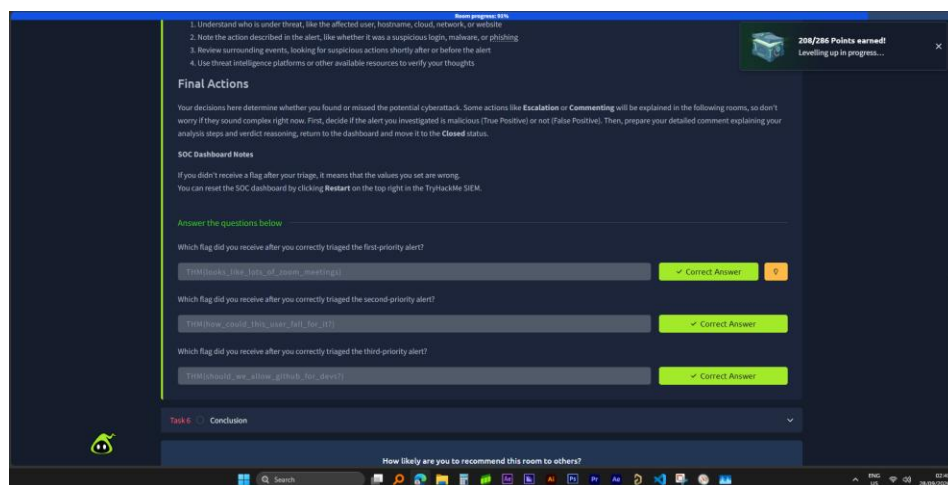


Figure 5 – SOC L1 Alert Triage: all three triage flags obtained correctly

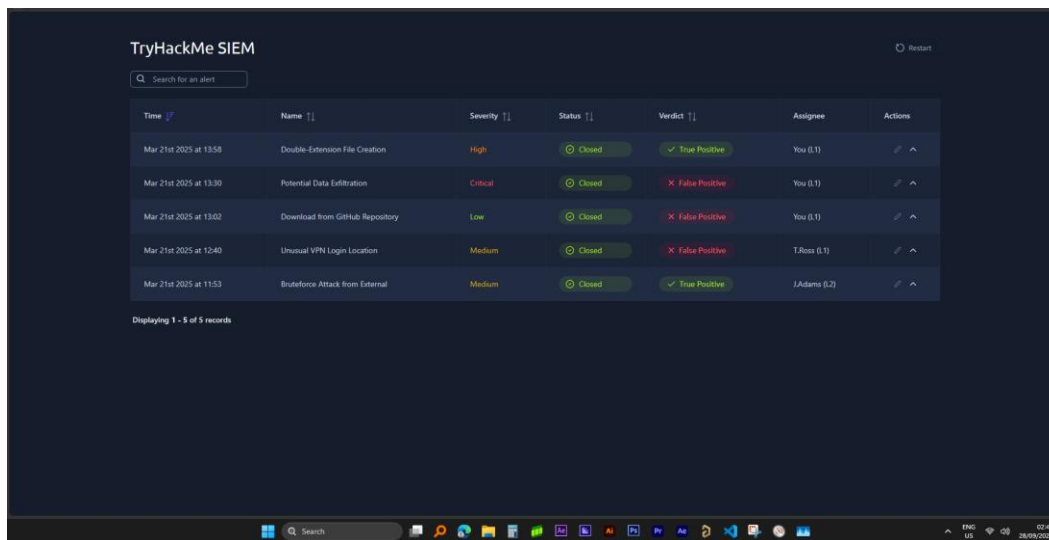


Figure 6 – TryHackMe SIEM dashboard showing all five alerts closed with correct verdicts: True Positive (×2) and False Positive (×3)

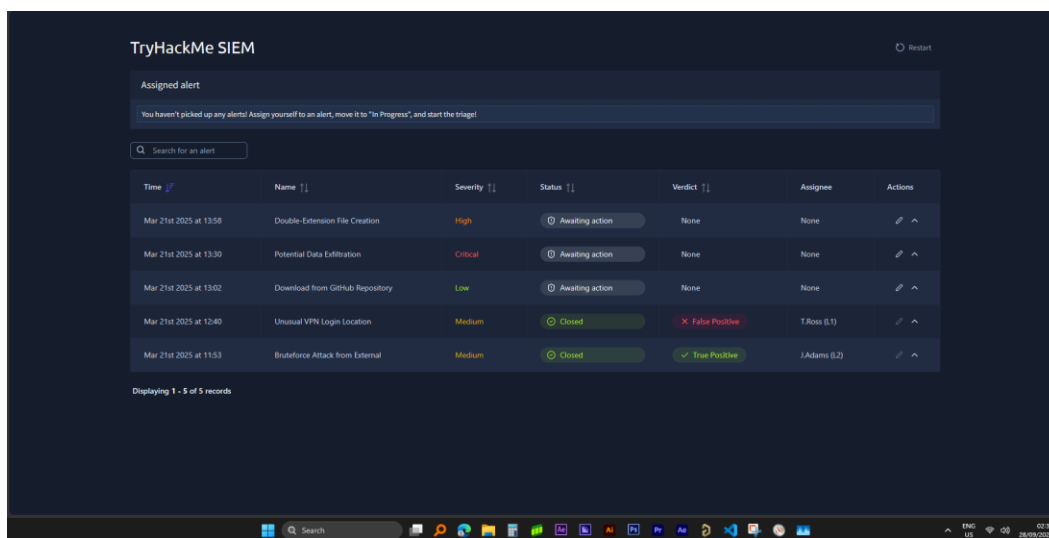


Figure 7 – TryHackMe SIEM at start of triage exercise: five alerts awaiting action